

# OPAUDIT 016 — A 16-bit file with an 8-bit picture inside it

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| Audit ID       | 2026-09-14-016                                                                                                                                                                                                                 |
| Report date    | 2026-09-14                                                                                                                                                                                                                     |
| First observed | 2026-09-14, the first day a RAW file went through SNAP SLAPPER since the RAW handoff was built. Every edit in the month before was made from JPEGs.                                                                            |
| Theme          | SNAP SLAPPER is specified as a 16-bit non-destructive editor. It was 16-bit at the door and 8-bit in the room where the editing happens. The file it wrote said 16-bit; the picture inside had already been reduced to 8-bit.  |
| Surfaces       | SNAP SLAPPER’s compositor (layers, blends, masks, effects); its TIFF export; the RAW handoff to RawTherapee; the “open developed file” step.                                                                                   |
| State          | <b>BLEW UP / OPEN</b> Diagnosed, not fixed. The compositor has to be rebuilt off Pillow. No build carries the fix yet.                                                                                                         |
| Watched live?  | <b>The defect, yes.</b> A real RAW, developed to a real 16-bit TIFF, came out of the app as 8-bit data in a 16-bit container. The fix has not been built, so nothing about the fix has been watched.                           |
| Reporter       | Sean (put the first real RAW through it and asked why the app still showed an 8-bit engine) + Codex (diagnosed it, once asked plainly, and did not dress it up).                                                               |
| Related        | DING DONG BELL “Photo editor files opened in other editors” — the Photoshop difference test on that entry was run on an 8-bit source, so it proved the wrapper and not the bits; it stands, with that limit now written on it. |
| Publication    | Public. No exploitable detail. Anyone who used SNAP SLAPPER on a high-bit source is affected and should know.                                                                                                                  |

## 1. SUMMARY

RawTherapee develops a RAW to a real 16-bit TIFF and hands it to SNAP SLAPPER correctly. SLAPPER’s compositor is built on Pillow. Pillow has no multi-channel high-bit image mode, so the first thing the compositor does with a 16-bit image is convert it to 8-bit RGB. Every layer, blend, mask and effect then runs in 8-bit, and the result is written into a 16-bit TIFF. The container is honest; the contents are not. The original RAW was never touched — the editor is non-destructive at the source, and that held. But every edit of a high-bit image made through the app was quietly degraded. It went unnoticed for a month because the month’s work was all JPEGs: 8-bit in, 8-bit composite, 8-bit out, nothing to lose, nothing visibly wrong.

## 2. WHAT A HUMAN SAW

| Step | What happened |
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|                                                      |                                                                                                                                                                                                                      |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open a RAW in SLAPPER                                | The app refused to develop it and offered only to open the original in RawTherapee. The RAW handoff (SLAPPER's sliders driving RawTherapee, a developed 16-bit TIFF coming back) had to be built first.              |
| Handoff reported done — “6 RAW-focused tests passed” | The developed file would not open. The app tried to open the RawTherapee settings sidecar ( .pp3 ) as if it were the photograph. Six tests had passed without one of them opening a developed image in the live app. |
| Ask why the app still shows an 8-bit engine          | Codex diagnosed it: RawTherapee produces 16-bit, the Pillow compositor downgrades to 8-bit before any layer or effect runs. The “16-bit editor” claim was true at RAW decode and false at composite.                 |

### 3. ROOT CAUSE — THREE OF THEM

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**The compositor never left 8-bit.** Pillow can hold a single 16-bit channel or a float channel, but not 16-bit RGB. The whole editing stage therefore ran in 8-bit regardless of what it was handed. This is the defect that matters.

**Files identified by guessing their names.** The .pp3 -opened-as-a-photo error, and a separate preview that snapped back to a stale cached TIFF, are the same bug: the pipeline works out which cached file is current by building a path string (add .tif , or open the .pp3 ) instead of tracking each file it makes by an assigned identity. Guess a filename and sooner or later you open the wrong file.

**Done was claimed twice for a path nobody had run.** Completion was reported — first for the handoff, then for RAW support — without a real RAW ever going through the live app. Green tests on the one workflow the feature exists for were taken as proof it worked. They were proof the tests ran.

### 4. EVIDENCE AT DIAGNOSIS

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One real RAW, developed by RawTherapee to a 16-bit TIFF (confirmed at that step). SLAPPER's compositor entry point converting to Pillow RGB before layer work. Export writing a 16-bit TIFF from that 8-bit working image. The RAW test suite: six passing tests, none of which opened a developed file in the app. A month of exports, all from JPEG sources, all internally consistent and therefore silent.

### 5. WHAT WE DID

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Nothing shipped yet. This entry is the disclosure. The rework, in one engine pass:

**Work in float32 internally; quantise to 16-bit only on export.** Not 16-bit integer — that still clips highlights on the first curve. Float keeps the headroom that makes highlight recovery from a RAW possible at all.

**Move colour compositing off Pillow.** Pillow stays for 8-bit legacy reading and writing only.

**Assigned identities for every file the pipeline makes.** A typed registry where a settings sidecar can never be opened as an image and a preview proxy can never become an edit source. Same pass as the bit-depth work, because the two bugs share the cause.

## 6. VERIFIED — AND WHAT REMAINS

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Verified: the defect, on a real RAW, on 2026-09-14. Not verified: anything else. To close this: put a real RAW through the rebuilt app, export a 16-bit TIFF, and prove the bits — a difference against the RawTherapee-developed TIFF, or a histogram that shows more than 256 levels per channel — not the declared depth in the header. Until that is watched, this stays open and SNAP SLAPPER is an 8-bit editor for anything above 8-bit, whatever the spec says.

## 7. WHAT WAS DAMAGED, AND WHAT WAS NOT

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**Safe** Original RAW files. Never modified.

**Safe** The month of JPEG edits. 8-bit sources composited in 8-bit lost nothing they had to give.

**Degraded** Any edit of a high-bit image made through the app: 8-bit data inside a 16-bit file. The RAW path — the path the tool exists for.

Caught on the first day of real RAW use rather than after a month of masters had gone through it.

## 8. EYEBALL NEXT

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The DING DONG BELL Photoshop entry: re-run the difference test from a RAW-developed source once the rework lands, because the existing pass was on an 8-bit source. Then every other place the app promises a bit depth — the export dialog, the file info panel, the LEWKS pipeline — gets the same prove-the-bits check.

## 9. THE DOCTRINE THIS INCIDENT CREATED

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*A container is not its contents. A 16-bit file can hold an 8-bit picture. Checking that the output is a 16-bit TIFF proves the wrapper; prove the bits. And “done” on a feature is watching a real file go through the live app and come out right — a green test suite on a path nobody has walked certifies nothing.*